Deep Ensemble CNN Method Based on Sample Expansion for Hyperspectral Image Classification

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***Abstract*— With the continuous progress of computer deep learning technology, convolutional neural network (CNN), as a representative approach, provides a unique solution for hyperspectral image (HSI) classification. However, the parameters of CNN cannot be well-tuned when the number of training samples is insufficient, resulting in unsatisfactory classification performance. To tackle the thorny problem, a deep ensemble CNN method based on sample expansion for HSI classification is studied in this article. In particular, spatial information is first extracted and fused with original spectral bands to help classifiers obtain discriminant spectral–spatial features. Then, we use the pixel-pair feature (PPF) to expand the number of training samples so that the parameters of CNN structure can be fully trained. In addition, deep ensemble CNN is employed in this article, enabling the trained model to obtain better generalization ability and more robust classification results. Ultimately, the proposed method is applied to classify four widely used hyperspectral datasets. Experimental results show that the studied approach yields higher classification accuracy than some CNN-based methods even under the condition of small-size training set.**

***Index Terms*— Ensemble convolutional neural network (CNN), hyperspectral image (HSI) classification, pixel-pair feature (PPF), small samples, spectral–spatial fusion (SSF).**

I. Introduction

HYPERSPECTRAL imaging spectrometers can capture dozens to hundreds of narrow bands in the same region of the Earth's surface, enhancing the expression ability of hyperspectral data for ground objects. Therefore, hyperspectral remote sensing technology has been widely applied to fine agriculture, military, soil investigation, environmental monitoring, and so on [1]–[3]. Hyperspectral image (HSI) classification has always been one of the critical contents in the process of hyperspectral data analysis [4]. It makes full use of the data characteristics and advantages of HSI to complete the process of assigning the unique category identification to the target pixel. Accurate classification results are the premise for the wide applications of hyperspectral remote sensing.

However, there are several challenges in the task of HSI classification.

1) *Hughes Phenomenon:* There are a maximum number of features for a given number of samples. When the number of features actually used exceeds this maximum, the classifier's performance is not improved but degraded [5], [6].
2) *Limited Number of Labeled Samples:* On the one hand, enough expert knowledge and human participants are required in the process of HSI annotation. On the other hand, one pixel represents the surface feature information of several meters due to the low spatial resolution of HSI data. Therefore, the proportion of labeled samples in the whole dataset is very small. However, the number of labeled samples occupies a vital position in the performance of classification algorithms.
3) *The Phenomena of "Same Object With Different Spectrum" and "Foreign Objects With the Same Spectra" [7], [8]:* As the ground objects contained in HSIs are of various types and complex backgrounds, the same objects may show relatively different spectral curves. In addition, two different ground objects, affected by light and density, may present similar or even identical spectral characteristics.

To deal with the complex issues of high dimensionality of features and low-sized training datasets, reducing spectral channel dimensions and fusing spatial elements are two traditional methods. Also, classifiers suitable for small samples are studied. With the purpose of reducing the dimension of spectral channels, feature transformation [9] and band selection [10] are two effective methods to extract spectral features. Considering the spatial and spectral features of ground objects comprehensively is also an efficient method for reducing the impact of Hughes phenomenon. Introducing spatial contents is conductive to reduce the uncertainty of samples and control the impact of noise on classification results. Commonly used small sample HSI classification approaches include the

Manuscript received February 12, 2022; revised April 6, 2022; accepted May 30, 2022. Date of publication June 15, 2022; date of current version June 27, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 61772397 and Grant 12005169, in part by the Natural Science Basic Research Program of Shaanxi under Grant 2021JC-23, and in part by the Science Technology and Development Project of Yulin Science and Technology Bureau under Grant CXY-2020-094. *(Shuxian Dong and Wei Feng contributed equally to this work.) (Corresponding authors: Wei Feng; Yinghui Quan.)*

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Digital Object Identifier 10.1109/TGRS.2022.3183189

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nearest neighbor and logistic regression analysis. Also, the support vector machine (SVM) [11] method shows superior performance.

Recently, machine learning-based approaches have made a breakthrough in computer vision. Also, deep learning is a further extension in the field of machine learning. It has been spawned many types of deep network models, including deep belief network (DBN), stacked autoencoder (SAE), recurrent neural network (RNN), and convolutional neural network (CNN). Among the numerous deep learning-based networks, CNN is the most representative and popular framework and has been proven effective in HSI classification [12]. Wei *et al.* [13] first explored the use of CNNs for HSI classification. The architecture containing five layers was employed to extract features directly from the spectral bands, yielding better classification results than SVM and some conventional deep learning-based approaches. After that, numerous CNN-based approaches have been developed for HSI classification [14]–[18]. Chen *et al.* [14] proposed an automatic CNN framework for the HSI classification task, in which the optimal architecture is found using a gradient descent-based search algorithm. He *et al.* [15] applied transfer learning to CNN-based HSI classification so as to short the consumption of training time. Mou and Zhu [16] designed a spectral attention module to enhance the representative capacity and interpretability of CNN architecture.

Since HSIs not only provide high-resolution spectral signatures but also contain rich spatial information, spectral–spatial features are considered to improve the classification performance of CNNs for HSI scenes [19]–[26]. Cao *et al.* [19] applied Markov random field (MRF) to smooth the predicted labels of CNN. In [20], a novel synergistic convolutional neural network (SyCNN) combining 2-D and 3-D CNNs in feature learning was proposed for accurate HSI classification, which acquired higher classification accuracy than some excellent methods that use 2-D/3-D CNNs. Zhu *et al.* [21] proposed a residual spectral–spatial attention network for the HSI classification task. Hong *et al.* [22] fused graph convolutional networks (GCNs) and CNNs to break the performance bottleneck of a single model. All of those CNN-based methods obtained appealing performance in HSI classification. However, most of them are based on sufficient training of the networks using a large number of labeled samples [27], [28]. Classification models based on CNN can easily fall into the overfitting problem when the number of labeled samples is insufficient. Therefore, how to use the powerful feature extraction ability of CNN to achieve HSI small sample classification is a challenging problem in remote sensing [29], [30].

During the last decades, ensemble learning-based approaches are also extensively investigated in HSI classification [31], [32]. The underlying idea of ensemble learning is that even if an individual classifier generates a wrong prediction, other classifiers in the ensemble framework can correct the error. Therefore, ensemble learning-based methods can produce more robust classification results. There are a series of shallow ensemble learning methods: ensemble frameworks based on k-nearest class collaborative representation [33], ensemble SVM [34], [35], ensemble extreme learning machine (ELM) [36], [37], random forest [38], rotation forest [39], [40], and so on. For example, Waske *et al.* [34] constructed an ensemble framework by integrating SVM and random subspace (RS) strategy. Also, it was proven to acquire higher classification accuracy than the standard SVM classifier. Apart from the shallow ensemble classifiers mentioned above, numerous deep ensemble learning-based approaches have also been comprehensively studied in recent years [41]–[45]. In [41], SAE was applied to assure the diversity between different base classifiers. Gong *et al.* [42] proposed a DBN-based ensemble framework. In addition, Chen *et al.* [43] combined CNN with ensemble learning. All of those methods obtain considerable classification performance when processing hyperspectral and multisource datasets.

Based on the fact that both shallow and deep ensemble learning are conducive to improving the classification performance of classifiers for HSI, ensemble learning is considered in this article. In particular, we explore a deep ensemble CNN structure based on sample expansion to alleviate the overfitting problem and enhance the stability of CNN-based classifiers. First, the spatial information is extracted from the original HSIs and spectral–spatial fusion (SSF) is obtained by channels superposition. Second, the pixel-pair feature (PPF) [46] is applied on the acquired spectral–spatial pixels to expand the number of labeled training samples. Third, an ensemble CNN framework is constructed to extract discriminant features from the spectral–spatial pixel pairs. Also, the final predicted labels are decided by voting the classification results of multiple individual classifiers in the ensemble CNN structure. The main contributions of this article are summarized as follows.

1) We propose a novel ensemble CNN method based on PPF for HSI classification, which aims to enhance the classification performance of CNN framework, especially under the condition of small samples.
2) To acquire more accurate classification results, we adopt the gray-level co-occurrence matrix GLCM) to extract spatial information from the original spectral bands. Then, the spectral information and the extracted spatial information are fused by bands superposition.
3) To ensure that the CNN model can be sufficiently trained, the PPF algorithm is applied on SSF information to realize the expansion of samples.
4) Given the superior performance of ensemble-SVM compared with SVM [34], ensemble CNN is explored to further improve the generalization ability and stabilize the classification performance. Moreover, the impact of ensemble size on the HSI classification accuracy is discussed on four public datasets.

The remainder of this article is structured into five sections. A brief overview of the related works is presented in Section II. The detailed descriptions of the proposed deep ensemble CNN based on PPF and SSF are given in Section III. Section IV reports the experimental results and analysis on four public datasets. The discussion and conclusion are involved in Sections V and VI, respectively.

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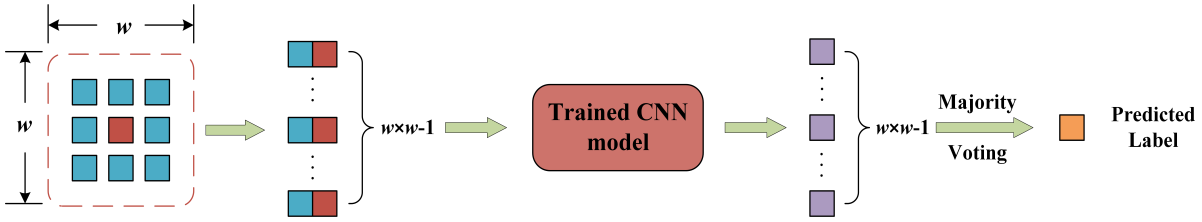


Fig. 1. Testing process based on the PPF algorithm.

II. Related Works

A. Theory of CNN Framework

CNN is a hierarchical structure composed of a deep stack of convolutional layers [47]. Compared with other neural networks, CNN has the following unique factors: 1) local connection and weight sharing can significantly reduce the connections between network neurons and the network complexity; 2) the parameter sharing in the same layer is beneficial to the parallel computation of the network, which can save a lot of training time under the condition of substantial image data and deep network structure; 3) CNN can process the input 2-D or 3-D images directly, avoiding the step of image structure destruction such as compression and stretching; and 4) feature extraction and feature classification are trained as a whole in a model, avoiding the inefficient manual feature extraction process.

A well-designed CNN framework usually comprises several convolutional layers, pooling layers, and fully connected layers. The convolutional layer has the characteristics of local receptive field and weight sharing among the same layers, which is the critical structure of CNN. Each convolution layer contains a certain number of convolution cores, which acts as filters in signal processing. The convolution can be formulated as

$$X_j^l = f\left(\sum_{i=1}^{M} X_i^{l-1} * k_{ij}^l + b_j^l\right) \tag{1}$$

where X_j^l is the jth feature map of the lth layer, X_i^{l-1} is the ith feature map of the $(l-1)$th layer, M is the number of feature maps in the $(l-1)$th layer, $*$ is the convolution operation, k_{ij}^l represents the convolution kernel weight between the ith feature map of the $(l-1)$th layer and the jth feature map of the lth layer, and b_j^l denotes the offset of the jth feature map of the lth layer. k_{ij}^l and b_j^l are two trainable parameters that can be fine-tuned using the backpropagation algorithm. $f(\cdot)$ is an activation function used to add nonlinear factors to the model and improve the expression ability of CNN. Rectified linear unit (ReLU) activation function is a commonly used and effective activation function. The advantage of this function is that there is no gradient saturation problem and the calculation speed is relatively fast, conducive to saving network training time. The ReLU function can be formulated as

$$\text{ReLU} = \max(0, X). \tag{2}$$

The pooling layer is mainly used to reduce the dimension of feature maps and compress the number of parameters [48]. Therefore, it can speed up the calculation process and alleviate the overfitting problem. For the ith input feature map, the output of linear downsampling is given as follows:

$$X_i^l = \text{down}\left(X_i^{l-1}\right) \tag{3}$$

where X_i^{l-1} and X_i^l denote the ith input and output feature maps of the lth layer, respectively. down$(\cdot)$ represents the pooling function, which generally adopts max pooling or average pooling.

The fully connected layer is usually placed after multiple convolutional or pooling layers. It maps the multidimensional features extracted from the previous layer to a vector label space to prepare for the subsequent classification task in image classification. Finally, the softmax function is applied to count the relative probabilities between different classes.

B. PPF Algorithm

To settle the scarcity of labeled pixels, Li *et al.* [46] proposed the PPF algorithm. In particular, on the training stage, a pixel-pair is expressed as $\mathbf{S}_{ij} = [\mathbf{x}_i, \mathbf{x}_j]$, where $\mathbf{x}_i$ and $\mathbf{x}_j$ are two pixels in the training set. The label of $\mathbf{S}_{ij} = [\mathbf{x}_i, \mathbf{x}_j]$ is determined by the following equation:

$$\text{Label}(\mathbf{S}_{ij}) = \begin{Bmatrix} c & \text{if Label}(\mathbf{x}_i)=\text{Label}(\mathbf{x}_j)=c \\ 0 & \text{else} \end{Bmatrix} \tag{4}$$

where $c = 1, 2, \ldots, C$, with C the number of categories in the HSI. Note that the PPF algorithm classifies the newly generated pixel-pairs into $C+1$ classes. Let n_c stand for the number of training samples in the cth class, and the total number of samples in the training set can be formulated as $N = \sum_{c=1}^{C} n_c$. As a result, the number of pixel-pairs in the cth class (represented as n_c') can be calculated as $n_c' = n_c \times (n_c - 1)$. Therefore, the number of pairs is $N' = \sum_{c=0}^{C} n_c'$, which is much larger than the original size N.

During the testing stage, pixel-pairs consist of the center pixel and its neighbor samples. The testing process based on the PPF algorithm is shown in Fig. 1. As the first step, a square window with size $w \times w$ is centered on the testing pixel. The second step is the construction of testing pixel-pairs using the center testing pixel and its surroundings. After the $w \times w - 1$ pixel-pairs fed into the trained CNN model, there are $w \times w - 1$ predicted labels. Also, the final classification label is decided via carrying out a voting strategy on the $w \times w - 1$ labels.

III. Proposed Method

The detailed theory of the proposed method is described in this section. It mainly includes three parts: spatial features extraction and SSF, the CNN structure adopted in this article, and the detailed description of the studied E-CNN-PPF-SSF approach. The flowchart of the proposed method is shown in Fig. 2.

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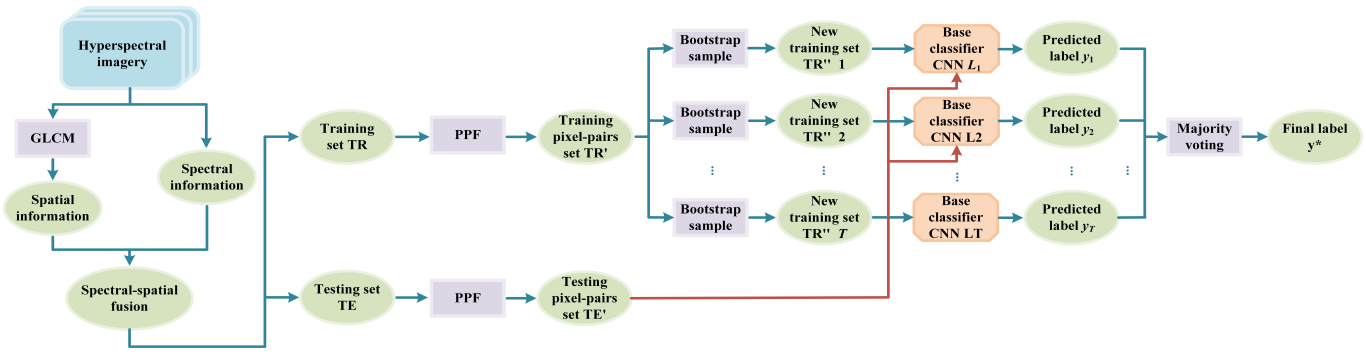


Fig. 2. Flowchart of the proposed E-CNN-PPF-SSF.

A. Spatial Features Extraction and SSF

Since numerous studies have demonstrated that classifiers that consider both spectral and spatial information of HSIs can obtain more accurate classification results than those only use spectral channels [14], SSF is developed in this article. GLCM is one of the typical spatial representation measures, which describes the position correlation between pixels with the same gray value or similar gray value [49]. Suppose that the size of HSI matrix R is $L_x \times L_y \times d$, where L_x and L_y denote the number of rows and columns of HSI, respectively, and d is the number of spectral bands. First, the value of each band of matrix R is quantized into N_g levels to obtain the new matrix R'

$$R'(x, y, d) = \left\lfloor (N_g - 1 + 0.9999) \frac{R(x, y, d) - \min(R(:, :, d))}{\max(R(:, :, d)) - \min(R(:, :, d))} \right\rfloor \quad (5)$$

where $\lfloor\rfloor$ is the round down operation; $R(x, y, d)$ and $R'(x, y, d)$ stand for the value of matrix R and R' in the row x, column y, and band d; and $\min(R(:, :, d))$ and $\max(R(:, :, d))$ represent the minimum and maximum values of the dth spectral channel in the matrix R, respectively.

Second, GLCM is adopted to extract the spatial information from the quantized matrix R'. For each center pixel (x, y), two sliding windows, named basic window w_1 and transformed window w_2, are required in the calculation of GLCM. Let i and j denote the gray tone of pixels in the basic window w_1 and transformed window w_2, $0 \le i, j \le N_g - 1$, d_x and d_y represent the number of rows and columns between w_1 and w_2, and GLCM (denoted by $G(i, j, d_x, d_y)$) can be formulated by the following mathematical equation:

$$G(i, j, d_x, d_y) = \sharp\{[(m, n), (m', n')], (m, n) \in w_1, (m', n') \in w_2 | \times f(m, n) = i, f(m', n') = j, m' - m = d_x, n' - n = d_y\} \quad (6)$$

where $\sharp$ represents the number of elements in the collection, (m, n) and (m', n') denote the pixel coordinates in the basic window w_1 and transformed window w_2, respectively, and $f(m, n)$ and $f(m', n')$ are the gray values corresponding to the point coordinates (m, n) and (m', n'), respectively.

Then, the GLCM $G(i, j, d_x, d_y)$ is normalized to probability matrix $G'(i, j, d_x, d_y)$

$$G'(i, j, d_x, d_y) = \frac{G(i, j, d_x, d_y)}{\sum_i \sum_j G(i, j, d_x, d_y)}. \quad (7)$$

In general, due to the large dimension of GLCM ($N_g \times N_g$), it is inappropriate to be directly used for the representation of spatial information. Based on the calculated GLCM, Haralick *et al.* [50] developed 14 statics to described texture properties of images. In this article, the mean static [denoted by $f'(x, y)$] is applied to occupy the spatial distributions of HSI

$$f'(x, y) = \sum_{i=0}^{N_g - 1} \sum_{j=0}^{N_g - 1} i \cdot G'(i, j, d_x, d_y). \quad (8)$$

It should be noted that $f'(x, y)$ is set to 0 when there are pixels in w_1 or w_2 but not in the matrix R'.

There are three popular fusion strategies: decision fusion, probability fusion, and stacking fusion. Among them, stacking fusion is the most common fusion scheme and has been intensively used for the classification task [51], [52]. Therefore, the spatial information captured from each band by the GLCM is stacked with the original spectral bands to construct SSF features in our method.

B. CNN Framework

The CNN architecture adopted in this article is presented in Fig. 3, which contains eight convolutional layers, three max-pooling layers, and two fully connected layers. In the convolution layer, features in the pixel-pair are automatically extracted by convolution kernels. It should be noted that the number of convolution kernels is equal to the number of feature maps in the next layer. Supposing that the input dimensionality is d_1 and the convolution kernel size is k_1, $d_1 - k_1 + 1$ features are produced after the convolution operation. Notably, the $d_1 - k_1 + 1$ features are more abstract and complex than the input features. The pooling layer is used to downsample the features from the convolution layer. After this step, the number of features is continuously compressed and the number of parameters is significantly reduced. Assuming that the input dimensionality is d_2 and the kernel size is k_2,

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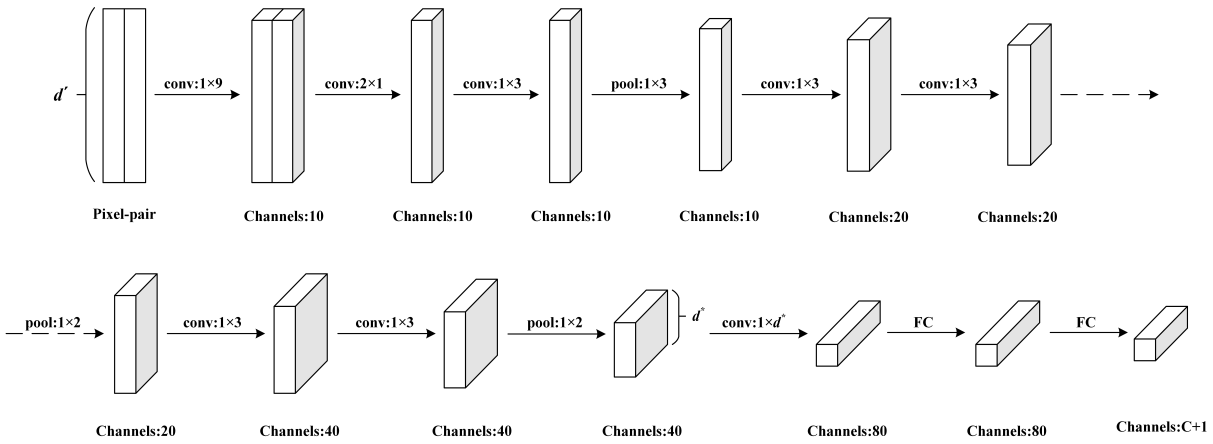


Fig. 3. CNN structure used in the proposed method.

d_2/k_2 features are generated after the pooling operation. This article applies the max-pooling function to take the maximum value of all elements in the pooling window as the eigenvalue of the subsampled feature map. Before the output layer of CNN, the learned distributed feature representation is mapped to the sample label space by two fully connected layers, realizing the explicit presentation of classification. It also should be noted that the ReLU function is implemented after each convolutional and fully connected layer, which aims to add nonlinear factors between different layers and improve the expression ability of CNN to the actual model. In the end, the output vector of the last fully connected layer is normalized to a category probability distribution by utilizing the softmax function.

C. Ensemble CNN Based on PPF and SSF

The steps of the proposed E-CNN-PPF-SSF approach are summarized in Algorithm 1. Also, detailed implementation processes of the studied approach are given as follows.

1) After capturing the spatial features $\mathbb{X}'$ from each band of the original HSI $\mathbb{X}$ using GLCM, spectral–spatial information $\mathbb{X}_F$ is obtained by the stacking strategy. Assuming that the number of spectral channels of a pixel is d, the number of features of a spectral–spatial pixel is transformed to $2d$ after the fusion operation.
2) The pixels in $\mathbb{X}_F$ are randomly divided into the training set **TR** and the testing set **TE** according to a certain proportion.
3) To augment the number of labeled training samples, the PPF is adopted on the training set **TR** and testing set **TE**. According to the PPF algorithm, each training pixel-pair is composed of two pixels that are randomly sampled from the original training set **TR**. The label of a training pixel-pair is determined using (4). For each testing pixel, testing pixel-pairs are constructed with their surrounding pixels. Also, testing pixel-pairs will be used to predict the category of the testing pixel via a voting strategy.
4) After obtaining training and testing pixel-pairs, the bootstrap sample technique is performed on the training pixel-pairs set $\mathbf{TR}'$ to construct various training sets $\mathbf{TR}''$. In particular, the number of pixel-pairs in $\mathbf{TR}''$ is equal to which in $\mathbf{TR}'$.
5) The CNN framework adopted in the proposed approach is presented in Fig. 3. First, the new training pixel-pairs

Algorithm 1 Ensemble CNN Based on PPF and SSF

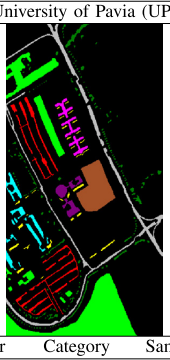
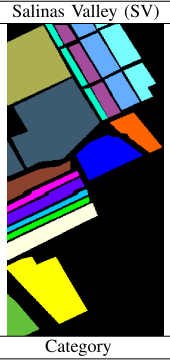
Require: $\mathbb{X}$: hyperspectral data; C: number of categories; d: number of spectral bands of $\mathbb{X}$; N: number of training samples; L: sub-classifier; T: number of sub-classifiers in the ensemble framework (i.e. ensemble size); y: prediction of base classifier; $\mathbf{E} = \varnothing$: an ensemble.

Process:
Extract the spatial features from the hyperspectral dataset $\mathbb{X}$ and obtain the spectral-spatial fusion information by stacking the original spectral bands with the captured spatial features
Acquire the training set **TR** by randomly selecting N pixels from the spectral-spatial dataset $\mathbb{X}_F$, construct the testing set **TE** by the remaining pixels
Obtain the training pixel-pairs set $\mathbf{TR}'$ and testing pixel-pairs set $\mathbf{TE}'$ by the PPF algorithm
for $t = 1 : T$ **do**
  Bootstrap sample pixel-pairs from $\mathbf{TR}'$, forming a new training set $\mathbf{TR}''$
  Training phase:
  Train a sub-classifier CNN L_t using the new training set $\mathbf{TR}''$
  Save the trained model L_t
  Testing phase:
  Generate prediction y_t by feeding the testing pixel-pairs set $\mathbf{TE}'$ into the trained CNN model L_t
end for

Ensure:
Ensemble CNN: $\mathbf{E} = \{L_1, L_2, \ldots, L_T\}$
Final prediction: $y^* = argmax \sum_{t=1}^{T} 1(y_t = c), c = 1, 2, \ldots, C$

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TABLE I
HSI DATA INFORMATION

University of Pavia (UP)			Salinas Valley (SV)			Indian Pines (IP)			Kennedy Space Center (KSC)		
											
Color	Category	Samples	Color	Category	Samples	Color	Category	Samples	Color	Category	Samples
	Asphalt	6631		Brocoli_green_weeds_1	2009		Alfalfa	46		Scrub	761
	Meadows	18649		Brocoli_green_weeds_2	3726		Corn-notill	1428		Willow swamp	243
	Gravel	2099		Fallow	1976		Corn-mintill	830		CP hammock	256
	Trees	3064		Fallow_rough_plow	1394		Corn	237		Slash pine	252
	Metal sheets	1345		Fallow_smooth	2678		Grass-pasture	483		Oak/Broadleaf	161
	Bare soil	5029		Stubble	3959		Grass-trees	730		Hardwood	229
	Bitumen	1330		Celery	3579		Grass-pasture-mowed	28		Swamp	105
	Bricks	3682		Grapes_untrained	11271		Hay-windrowed	478		Graminoid marsh	431
	Shadows	947		Soil_vinyard_develop	6203		Oats	20		Spartina marsh	520
				Corn_senesced_green_weeds	3278		Soybean-notill	972		Cattail marsh	404
				Lettuce_romaine_4wk	1068		Soybean-mintill	2455		Salt marsh	419
				Lettuce_romaine_5wk	1927		Soybean-clean	593		Mud flats	503
				Lettuce_romaine_6wk	916		Wheat	205		Water	927
				Lettuce_romaine_7wk	1070		Woods	1265			
				Vinyard_untrained	7268		Buildings-Grass-Trees-Drives	386			
				Vinyard_vertical_trellis	1807		Stone-Steel-Towers	93			
Total Number		42776	Total Number		54129	Total Number		10249	Total Number		5211

in the training set $\mathbf{TR}''$ are fed into the framework, obtaining a trained CNN classifier L. Then, the classifier is applied to classify the testing pixel-pairs set $\mathbf{TE}'$, acquiring an individual prediction result y.

6) Steps 4) and 5) are carried out T times, either serial or parallel, with T representing the ensemble size, generating the ensemble CNN model $\mathbf{E} = \{L_1, L_2, \ldots, L_T\}$ and T prediction results.
7) Since there is no need to assume any prior knowledge of the performance of subclassifiers, the majority voting is considered to be the most straightforward and the simplest decision fusion strategy [35], [43], [53]–[56]. Therefore, we apply the majority voting to determine the final predicted label y^* by combining T classification outputs

$$y^* = \text{argmax} \sum_{t=1}^{T} 1(y_t = c), \quad c = 1, 2, \ldots, C \tag{9}$$

where 1 (statement) is equal to 1 when the statement is true and is equal to 0 when the statement is false.

IV. EXPERIMENTAL RESULTS

In this article, all experiments are implemented in a hardware environment constituted by an Intel Xeon Silver 4210R processor and a graphic processing unit (GPU) NVIDIA Quadro RTX 5000 with 16-GB memory. The platform is Python 3.7 in a Tensorflow library. Experiment data, experiment settings, evaluation methods, and classification performance are displayed in detail in the following.

A. Experiment Data

Four representative HSI images, including University of Pavia (UP), Salinas Valley (SV), Indian Pines (IP), and Kennedy Space Center (KSC), are adopted to verify the validity of the studied E-CNN-PPF-SSF method. These datasets are publicly accessible online http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes. The detailed descriptions of four HSIs are presented in Table I.

1) *UP*: The UP image was gathered by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor at the Pavia University, Italy. It has 610 × 340 pixels and the spatial resolution is 1.3 meters per pixel (mpp). After abandoning the original 13 noisy spectral bands, the 103 reserved bands are used in the experiment. There are nine categories and 42 776 labeled samples, which roughly include trees, asphalt, buildings, and painted metal objects.
2) *SV*: The scene was acquired by the Airborne Visible-Infrared Imaging Spectrometer (AVIRIS) sensor over SV, California. The spatial size of the dataset is 512 × 217, which contains 54 129 labeled pixels. It has a total of 16 classes and its spatial resolution is 3.7 mpp. After removing 20 water-absorbing bands, 204 spectral bands are reserved in our experiment.
3) *IP*: The scene was collected by AVIRIS sensor over the IP test site in Northwestern Indiana. There are 145 × 145 pixels and the spatial resolution is 20 mpp. Approximately 10 249 available pixels with

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13 categories are considered for classification purposes. In addition, 200 spectral bands are used by discarding 20 water-absorbing bands.

4) *KSC*: The data were also gathered by the AVIRIS instrument over the KSC, Florida, on March 23, 1996. The scene is of size 512×614 and the spatial resolution is 20 m. It contains 5122 labeled pixels with 13 classes. After removing water absorption and low signal-to-noise ratio (SNR) bands, 176 bands are adopted for the analysis.

B. Experiment Settings

To demonstrate the classification performance of the proposed E-CNN-PPF-SSF, the following experiments are considered.

1) *CNN [13]:* A typical CNN framework with five layers, which is the first attempt to classify HSIs using the CNN structure. Since the optimizer is not specified in [13], the standard stochastic gradient descent (SGD) with momentum is adopted in our experiment.
2) *R-PCA-CNN [18]:* A combination of CNN framework and randomized principal component analysis (R-PCA), which conducts the task of high-level features construction.
3) CNN-MRF [19]: A integration of CNN and MRF, spectral, and spatial information are fully considered in the process of HSI classification.
4) *GCN-CNN [22]:* A concatenation fusion scheme that integrates features extracted from CNNs and minibatch graph convolutional networks (miniGCNs) in an end-to-end trainable network.
5) *CNN-PPF [46]:* A sample expansion method using PPFs, which aims to ensure that the parameters of CNN can be well-tuned. Experimental results in [46] show that the CNN-PPF can significantly enhance the classification performance even when the number of training samples is limited.

C. Evaluation Methods

The performance measures adopted in the experiments are overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC). These indexes are calculated based on the confusion matrix $[P_{cc'}]$, where $[P_{cc'}]$ denotes the number of tasting pixels whose true label is c and predicted label is c'. $[P_{cc'}]$ can be formulated by the following equation:

$$P_{cc'} = \sum_{p=1}^{P} 1\left(\widehat{y}_p = c\right) 1\left(y_p^* = c'\right) \tag{10}$$

where P is the number of testing pixels and $\widehat{y}_p$ and y_p^* represent the true label and predicted label of the pth testing pixel, respectively.

1) OA refers to the true prediction rate

$$\mathrm{OA} = \frac{1}{P} \sum_{c=1}^{C} P_{cc}. \tag{11}$$

2) AA stands for the average accuracy of each category

$$\mathrm{AA} = \frac{1}{C} \sum_{c=1}^{C} \frac{P_{cc}}{\sum_{c'=1}^{C} P_{cc'}}. \tag{12}$$

3) KC represents the proportion of error reduction caused by classification and completely random classification

$$\mathrm{KC} = \frac{\frac{1}{P}\sum_{c} P_{cc} - \frac{1}{P^2}\left(\sum_{c'} P_{cc'}\right)\left(\sum_{c'} P_{c'c}\right)}{1 - \frac{1}{P^2}\left(\sum_{c'} P_{cc'}\right)\left(\sum_{c'} P_{c'c}\right)}. \tag{13}$$

D. Classification Performance

1) Experimental Analysis: The setting of parameters is essential to the performance of classification methods. Appropriate parameter setting can make the methods obtain satisfactory classification accuracy. In this section, we first analyze the complexity of CNN model. Then, the optimal values of the two parameters, initial learning rate and ensemble size, are evaluated in this section. Finally, an ablation experiment is conducted to explore the contribution of different factors in the proposed method to the improvement of CNN classification performance. Specifically, we randomly choose 50 samples from each class to construct the training set, and the testing set is composed of the remaining pixels. Since the number of samples in the classes Alfalfa, Grass-pasture-mowed, and Oats of the IP data is particularly small, we only use half of their samples to construct the training set.

a) Computational complexity analysis: Two widely used indexes, trainable parameters and floating-point operations (FLOPs), are applied to analyze the complexity of CNN model. The number of trainable parameters corresponds to the space complexity (occupied memory size), while the amount of FLOPs denotes the time complexity (length of time). The larger the values of these two parameters are, the more complex the network is. Table II shows the number of trainable parameters and FLOPs required when the CNN model is trained with four different hyperspectral datasets. It can be seen that due to the different number of categories and bands in the four considered HSIs, the number of trainable parameters and FLOPs required for CNN model training is also different. Also, the more the number of categories and bands, the more trainable parameters and FLOPs are required. Because the number of categories and bands of the SV dataset is the largest, the trained model with this scene requires the most significant number of trainable parameters and FLOPs.

b) Initial learning rate: The learning rate controls the speed at which the depth model converges to the optimal solution in the process of backpropagation [46]. If the learning rate is set too large, the loss function may directly cross the global best point, resulting in an oscillation over training epochs. On the contrary, the change speed of the loss function is plodding when the learning rate is set too small, making the loss function easy to be trapped in the local minimum or saddle point. In this experiment, the number of total training epochs is set to 900 and the learning rate decays every 300 epochs with a decay rate of 0.01. The ensemble size is 5. Classification accuracy of the proposed method with different learning rates is reported in Table III. Also, the best classification results for

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TABLE II
ANALYSIS OF THE CNN FRAMEWORK

Layer	Output dimensions	Trainable parameters (k) / FLOPs (k)			
		UP	SV	IP	KSC
Input:	$2 \times 2d \times 1$	-	-	-	-
Conv 1 (10 filters, size 1×9)-ReLU	$2 \times (2d-8) \times 10$	0.10/75.24	0.10/152	0.10/148.96	0.10/130.72
Conv 2 (10 filters, size 2×1)-ReLU	$1 \times (2d-8) \times 10$	0.21/81.18	0.21/164	0.21/160.72	0.21/141.04
Conv 3 (10 filters, size 1×3, with padding)-ReLU	$1 \times (2d-8) \times 10$	0.31/120.78	0.31/244	0.31/239.12	0.31/209.84
Max-pooling (size 1 × 3)	$1 \times \lceil \frac{2d-8}{3} \rceil \times 10$	0/1.98	0/4.02	0/3.93	0/3.45
Conv 4 (20 filters, size 1×3)-ReLU	$1 \times \lceil \frac{2d-14}{3} \rceil \times 20$	0.62/78.08	0.62/161.04	0.62/157.38	0.62/137.86
Conv 5 (20 filters, size 1×3)-ReLU	$1 \times \lceil \frac{2d-20}{3} \rceil \times 20$	1.22/150.04	1.22/314.60	1.22/307.34	1.22/268.62
Max-pooling (size 1 × 2)	$1 \times \lceil \frac{2d-20}{3} \rceil \times 20$	0/1.24	0/2.60	0/2.56	0/2.24
Conv 6 (40 filters, size 1×3)-ReLU	$1 \times \lceil \frac{2d-32}{6} \rceil \times 40$	2.44/140.36	2.44/304.92	2.44/300.08	2.44/261.36
Conv 7 (40 filters, size 1×3)-ReLU	$1 \times \lceil \frac{2d-44}{6} \rceil \times 40$	4.84/260.28	4.84/588.04	4.84/578.40	4.84/501.28
Max-pooling (size 1 × 2)	$1 \times \lceil \frac{2d-44}{12} \rceil \times 40$	0/1.12	0/2.48	0/2.40	0/2.08
Conv 8 (80 filters, size $1 \times \lceil \frac{2d-44}{12} \rceil$)-ReLU	$1 \times 1 \times 80$	44.88/89.68	99.28/198.48	96.08/192.08	83.28/166.48
Dense (80 filters)-ReLU	$1 \times 1 \times 80$	6.48/12.88	6.48/12.88	6.48/12.88	6.48/12.88
Dense ($C+1$ filters)-softmax	$1 \times 1 \times (C+1)$	0.81/1.66	1.377/2.822	1.377/2.822	1.134/2.324
Total	-	61.91/1014.520	116.877/2151.882	113.677/2108.672	100.634/1840.174

TABLE III
CLASSIFICATION ACCURACY OF THE PROPOSED METHOD WITH DIFFERENT LEARNING RATES

Data set	Initial learning rate = 0.001			Initial learning rate = 0.01		
	OA(%)	AA(%)	KC(×100)	OA(%)	AA(%)	KC(×100)
UP	92.97±0.93	94.50±0.20	90.72±1.20	**94.22±0.46**	**95.68±0.10**	**92.41±0.58**
SV	**94.69±0.22**	**96.80±0.16**	**94.08±0.24**	94.42±0.11	96.60±0.10	93.78±0.13
IP	**94.28±0.27**	97.15±0.70	**93.46±0.31**	94.01±0.15	**97.27±0.11**	93.16±0.17
KSC	99.01±0.18	98.65±0.19	98.89±0.20	**99.13±0.11**	**99.06±0.15**	**99.02±0.13**

the four HSIs are shown in bold. As presented in Table III, the optimal learning rates based on the classification accuracy are 0.01, 0.001, 0.001, and 0.01 for the UP, SV, IP, and KSC datasets, respectively.

c) Ensemble size T: To discuss the sensitivity of the parameter T to the classification accuracy of the proposed method and select the optimal parameters, experiments with dynamic ensemble size ($T = 5, 10, 20, 30, 40,$ and 50) are carried out. The average classification results with standard variations and time consumption based on four HSI images are presented in Table IV. We can see that, first, with the increase of ensemble size T, the classification accuracy increases, and then, the accuracy remains at a certain level. At the same time, the standard deviation decreases with the increase of ensemble size in most cases, indicating that a larger ensemble size is helpful to improve the stability of classification performance. In the experiment of UP scene, the proposed method obtains the highest OA, AA, and KC when the ensemble size is $T = 30$. Also, the optimal parameter is $T = 50$ for the SV, IP, and KSC datasets. However, it should not be neglected that the time consumption is proportional to the ensemble size due to the serial training of individual classifiers. Therefore, we choose $T = 10$ after making a compromise between classification accuracy and time consumption.

d) Ablation study: To analyze the contribution of different factors in the proposed method to the improvement of classification performance of CNN model, ablation experiments using the UP dataset are carried out in this section. In Table V, CNN indicates that only the CNN model (shown in Fig. 3) is adopted to classify the UP scene, CNN-PPF [46] represents the method combined with sample expansion and CNN, E-CNN-PPF stands for the CNN-PPF approach jointed with ensemble learning, CNN-PPF-SSF refers to the CNN-PPF algorithm integrated with spatial information, and E-CNN-PPF-SSF denotes the proposed method, which combines both ensemble architecture and spatial information with the CNN-PPF approach. For the CNN-based ensemble frameworks E-CNN-PPF and E-CNN-PPF-SSF, the ensemble size is set to 10. As shown in Table V, the classification accuracy of the CNN-PPF approach is obviously higher than that of CNN, indicating that the PPFs are effective and meaningful. When the CNN-PPF algorithm is compared with E-CNN-PPF and CNN-PPF-SSF, the classification results of both methods are better than that of CNN-PPF, suggesting that integrating spatial information or ensemble framework based on CNN-PPF is conducive to the improvement of CNN classification performance. Moreover, it is noted that the studied method E-CNN-PPF-SSF yields the best classification result, with the gains of 14.53%, 4.35%, 3.43%, and 1.04% over CNN, CNN-PPF, E-CNN-PPF, and CNN-PPF-SSF. Therefore, we can conclude that PPFs, extracted spatial information, and ensemble architecture all contribute to the enhancement of CNN classification performance.

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TABLE IV
CLASSIFICATION RESULTS ACHIEVED BY THE PROPOSED METHOD USING DIFFERENT ENSEMBLE SIZES

Data set	Index	Ensemble size					
		5	10	20	30	40	50
UP	OA(%)	94.08±0.58	94.22±0.46	94.20±0.44	**94.24±0.45**	94.23±0.45	94.22±0.43
	AA(%)	95.61±0.14	95.68±0.10	95.69±0.10	**95.69±0.09**	95.67±0.10	95.67±0.10
	KC(×100)	92.22±0.74	92.41±0.58	92.39±0.52	**92.43±0.57**	92.43±0.57	92.41±0.55
	time(h)	1.5	3.1	6.1	9.1	12.3	15.3
SV	OA(%)	94.69±0.24	94.69±0.22	94.78±0.24	94.87±0.23	94.90±0.22	**94.90±0.20**
	AA(%)	96.88±0.07	96.80±0.16	96.95±0.12	96.99±0.10	97.02±0.10	**97.02±0.09**
	KC(×100)	94.09±0.26	94.08±0.24	94.19±0.26	94.29±0.25	94.32±0.27	**94.32±0.22**
	time(h)	3.5	7.0	14.0	20.9	27.9	34.9
IP	OA(%)	94.07±0.31	94.28±0.27	94.42±0.22	94.48±0.24	94.48±0.27	**94.52±0.19**
	AA(%)	97.06±0.71	97.15±0.70	97.21±0.70	97.25±0.69	97.26±0.71	**97.28±0.67**
	KC(×100)	93.23±0.35	93.46±0.31	93.62±0.25	93.69±0.27	93.69±0.31	**93.74±0.22**
	time(h)	2.8	5.6	11.2	16.9	22.5	28.1
KSC	OA(%)	98.92±0.18	99.02±0.04	99.09±0.09	99.11±0.09	**99.11±0.04**	99.11±0.05
	AA(%)	98.90±0.24	99.00±0.10	99.08±0.09	99.08±0.10	99.09±0.07	**99.10±0.06**
	KC(×100)	98.78±0.20	98.90±0.04	98.98±0.10	99.00±0.10	99.00±0.05	**99.01±0.06**
	time(h)	2.3	4.5	9.0	13.5	18.0	22.5

TABLE V
ABLATION ANALYSIS OF THE PROPOSED METHOD USING THE UP DATASET

Method	CNN	CNN-PPF	E-CNN-PPF	CNN-PPF-SSF	E-CNN-PPF-SSF
OA(%)	79.69±1.45	89.87±0.58	90.79±0.78	93.18±0.84	**94.22±0.46**
AA(%)	85.98±1.10	92.91±0.24	93.13±0.43	94.93±0.48	**95.67±0.10**
KC(×100)	74.04±1.78	86.81±0.73	87.92±0.97	91.05±1.08	**92.41±0.58**

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE UP DATASET

Category	Train	Test	CNN [13]	R-PCA-CNN [18]	CNN-MRF [19]	GCN-CNN [22]	CNN-PPF [46]	**E-CNN-PPF-SSF**
Asphalt	50	6581	80.07	87.77	89.45	81.32	95.19	94.53
Meadows	50	18599	83.69	84.10	89.58	94.90	85.98	93.01
Gravel	50	2049	80.83	85.64	84.65	85.53	89.46	92.76
Trees	50	3014	90.87	93.65	96.23	94.75	90.98	96.42
Metal sheets	50	1295	99.10	100	99.83	99.88	99.91	100
Bare soil	50	4979	88.97	85.66	89.48	89.62	92.87	95.92
Bitumen	50	1280	91.47	95.64	91.06	90.42	94.58	97.11
Bricks	50	3632	81.34	80.89	81.41	78.95	87.36	92.31
Shadows	50	897	99.46	99.15	99.98	100	99.13	99.02
OA(%)			84.96	86.52	89.66	90.46	89.87	**94.22**
AA(%)			88.42	90.28	91.30	90.60	92.83	**95.68**
KC(×100)			80.54	82.68	86.56	87.41	86.81	**92.41**

2) Comparisons With the State-of-the-Art Methods: To demonstrate the effectiveness of the proposed E-CNN-PPF-SSF method, five state-of-the-art methods, CNN [13], R-PCA-CNN [18], CNN-MRF [19], GCN-CNN [22], and CNN-PPF [46], are compared in this section. The settings of those comparison methods have been described in Section IV-B. Consistent with Section IV-D1, we randomly select 50 pixels from each category to construct the training set. The initial learning rates used in the experiment are set to 0.01, 0.001, 0.001, and 0.01 for the UP, SV, IP, and KSC datasets, respectively. Also, the ensemble size T is set to 10.

Classification results of the six methods for the four HSI datasets are given in Tables VI–IX. Obviously, there are significant improvements over the five comparison methods for the four HSI scenes. Taking Table VIII for example, our studied approach has 18.70%, 18.32%, 13.83%, and 16.03% improvements over the CNN, R-PCA-CNN, CNN-MRF, and GCN-CNN, respectively. Moreover, the proposed method

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TABLE VII
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SV DATASET

Category	Train	Test	CNN [13]	R-PCA-CNN [18]	CNN-MRF [19]	GCN-CNN [22]	CNN-PPF [46]	**E-CNN-PPF-SSF**
Brocoli_green_weeds_1	50	1959	96.87	98.83	95.50	99.20	98.66	98.17
Brocoli_green_weeds_2	50	3676	98.15	98.89	99.07	99.95	99.09	99.54
Fallow	50	1926	98.17	96.98	88.45	97.38	99.17	99.90
Fallow_rough_plow	50	1344	99.78	98.57	97.68	99.51	99.87	98.54
Fallow_smooth	50	2628	95.94	97.26	99.48	98.55	95.30	99.19
Stubble	50	3909	99.39	98.09	99.62	99.98	99.57	99.09
Celery	50	3529	99.11	98.82	99.68	99.81	99.61	99.11
Grapes_untrained	50	11221	76.42	80.32	85.26	66.84	84.84	91.06
Soil_vinyard_develop	50	6153	98.88	97.98	95.43	99.85	98.50	97.47
Corn_senesced_green_weeds	50	3228	87.32	93.85	97.24	93.71	90.90	91.80
Lettuce_romaine_4wk	50	1018	93.48	97.15	98.55	98.41	99.27	97.66
Lettuce_romaine_5wk	50	1877	99.77	100	99.99	99.94	99.99	99.79
Lettuce_romaine_6wk	50	866	98.52	100	99.93	99.88	99.19	98.82
Lettuce_romaine_7wk	50	1020	93.20	96.82	99.39	98.37	94.71	97.45
Vinyard_untrained	50	7218	70.67	79.92	71.49	73.82	75.23	84.40
Vinyard_vertical_trellis	50	1757	98.81	96.56	98.70	97.94	98.93	96.85
OA(%)			89.22	92.62	91.48	88.72	92.14	**94.69**
AA(%)			94.03	95.63	95.34	95.20	95.80	**96.80**
KC(×100)			88.02	91.79	90.51	87.47	91.24	**94.08**

TABLE VIII
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET

Category	Train	Test	CNN [13]	R-PCA-CNN [18]	CNN-MRF [19]	GCN-CNN [22]	CNN-PPF [46]	**E-CNN-PPF-SSF**
Alfalfa	23	23	95.65	91.30	95.65	95.65	95.65	97.39
Corn-notill	50	1378	69.03	63.00	78.40	60.26	85.24	95.80
Corn-mintill	50	780	71.46	76.18	86.92	72.18	84.00	94.97
Corn	50	187	87.06	87.59	99.15	92.09	94.22	99.04
Grass-pasture	50	433	91.96	90.33	96.44	91.96	91.59	94.87
Grass-trees	50	680	92.74	97.14	93.79	98.59	98.15	100
Grass-pasture-mowed	14	14	85.71	100	100	97.14	82.86	97.14
Hay-windrowed	50	428	96.59	99.77	97.15	98.04	99.16	99.91
Oats	10	10	100	100	100	100	96.00	98.00
Soybean-notill	50	922	78.72	72.54	75.27	70.95	84.08	97.18
Soybean-mintill	50	2405	61.32	60.95	59.89	73.74	72.08	85.20
Soybean-clean	50	543	80.96	62.58	76.50	77.72	92.85	98.42
Wheat	50	155	99.61	100	98.96	99.87	100	100
Woods	50	1215	82.80	93.65	94.47	86.29	86.68	97.64
Buildings-Grass-Trees-Drives	50	336	65.65	86.67	98.27	83.10	94.35	98.03
Stone-Steel-Towers	50	43	95.35	100	100	100	100	100
OA(%)			75.58	75.90	80.45	78.25	85.00	**94.28**
AA(%)			84.66	86.36	90.68	87.35	91.06	**97.10**
KC(×100)			72.38	72.67	77.92	75.19	82.95	**93.46**

yields 9.28% higher accuracy than the sample extension method CNN-PPF due to the introduction of ensemble CNN, which enhances the classification performance by considering spectral and spatial information simultaneously and constructing diverse classifiers.

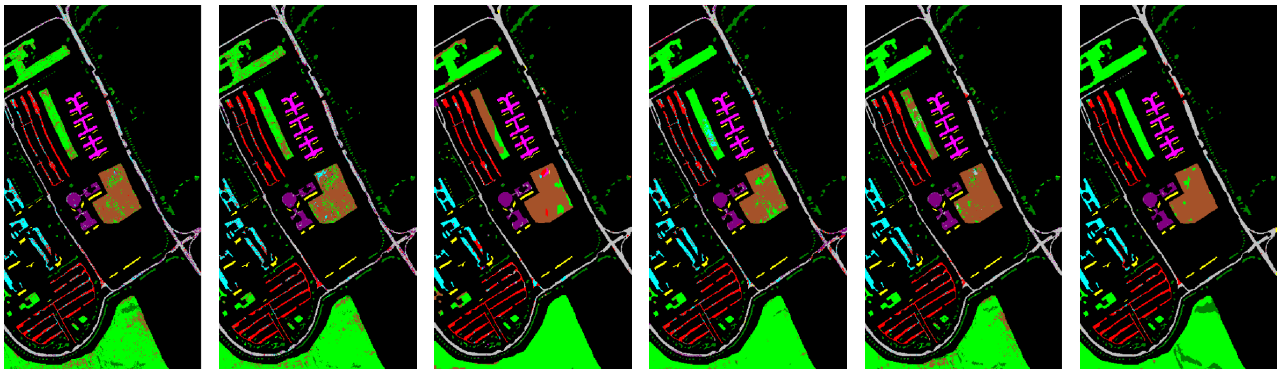
Figs. 4–7 show the qualitative maps of the six methods for the four HSIs. These maps correspond to the classification results presented in Tables VI–IX. From those classification maps, we can note that the proposed method acquires smoother and more detailed maps. For example, the area of Soybean-notill in Fig. 6(f) is less noisy than the reference methods.

To evaluate the classification performance of the six approaches and verify the sensitivity of those methods to the size of training set, 10, 20, 30, 40, and 50 pixels are randomly

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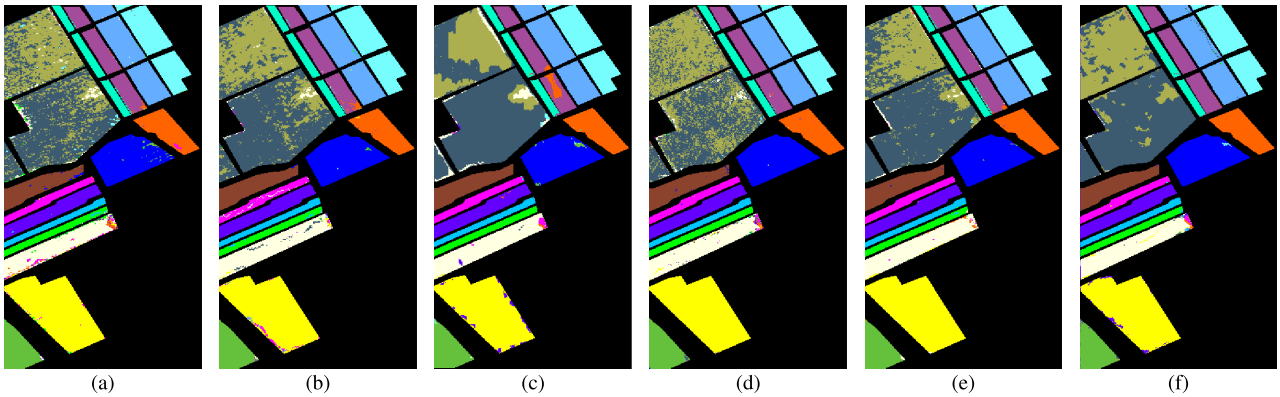
TABLE IX
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE KSC DATASET

Category	Train	Test	CNN [13]	R-PCA-CNN [18]	CNN-MRF [19]	GCN-CNN [22]	CNN-PPF [46]	**E-CNN-PPF-SSF**
Scrub	50	711	87.45	90.89	94.80	95.58	94.74	96.77
Willow swamp	50	193	86.94	94.51	98.96	97.51	91.40	97.62
CP hammock	50	206	90.00	87.09	98.54	95.53	96.41	98.54
Slash pine	50	202	67.23	56.73	81.68	55.64	88.32	96.44
Oak/Broadleaf	50	111	76.58	85.77	88.29	80.18	95.86	99.10
Hardwood	50	179	84.36	83.46	96.55	92.18	98.55	100
Swamp	50	55	94.18	98.91	100	91.64	100	100
Graminoid marsh	50	381	92.49	96.71	97.64	95.49	97.64	99.63
Spartina marsh	50	470	96.89	100	99.36	97.36	97.53	99.96
Cattail marsh	50	354	95.48	98.70	98.31	94.69	99.55	99.66
Salt marsh	50	369	96.91	99.78	99.19	99.51	98.92	99.73
Mud flats	50	453	92.49	95.23	99.34	89.71	97.53	99.56
Water	50	877	99.98	99.98	100	99.09	100	100
OA(%)			92.04	93.97	97.33	93.85	97.16	**99.02**
AA(%)			89.31	91.37	96.37	91.09	96.65	**99.00**
KC($\times$100)			91.08	93.24	97.00	93.11	96.80	**98.90**



(a) (b) (c) (d) (e) (f)

Fig. 4. Classification maps obtained by six methods on the UP dataset. (a) CNN. (b) R-PCA-CNN. (c) CNN-MRF. (d) GCN-CNN. (e) CNN-PPF. (f) E-CNN-PPF-SSF.



(a) (b) (c) (d) (e) (f)

Fig. 5. Classification maps obtained by six methods on the SV set. (a) CNN. (b) R-PCA-CNN. (c) CNN-MRF. (d) GCN-CNN. (e) CNN-PPF. (f) E-CNN-PPF-SSF.

sampled from each category for the training of classifiers. Notably, the number of pixels in the classes Alfalfa, Grass-pasture-mowed, and Oats of the IP image is particularly small, and we only chose half of their samples to formulate the training set when the number of training samples is more than half of the total number of samples. OAs, AAs, and KCs for various classifiers under each condition are shown in Figs. 8–11. In most cases, the six methods achieve higher classification accuracy when the training set size increases. Also, the proposed E-CNN-PPF-SSF method acquires better

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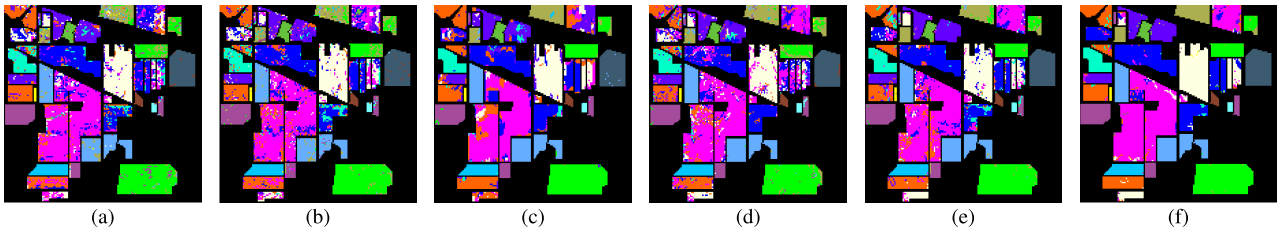


Fig. 6. Classification maps obtained by six methods on the IP dataset. (a) CNN. (b) R-PCA-CNN. (c) CNN-MRF. (d) GCN-CNN. (e) CNN-PPF. (f) E-CNN-PPF-SSF.

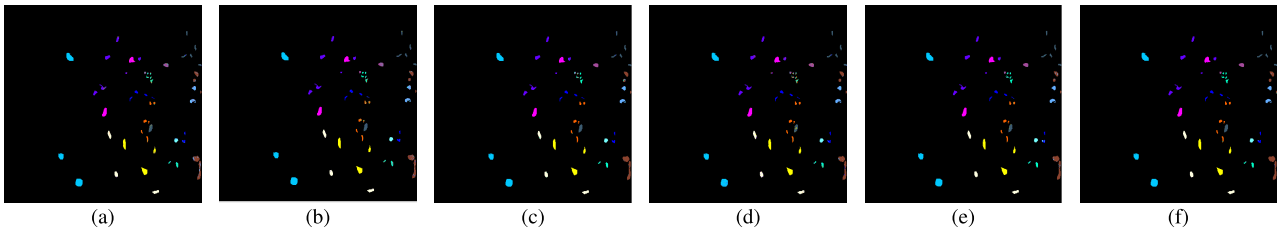


Fig. 7. Classification maps obtained by six methods on the KSC dataset. (a) CNN. (b) R-PCA-CNN. (c) CNN-MRF. (d) GCN-CNN. (e) CNN-PPF. (f) E-CNN-PPF-SSF.

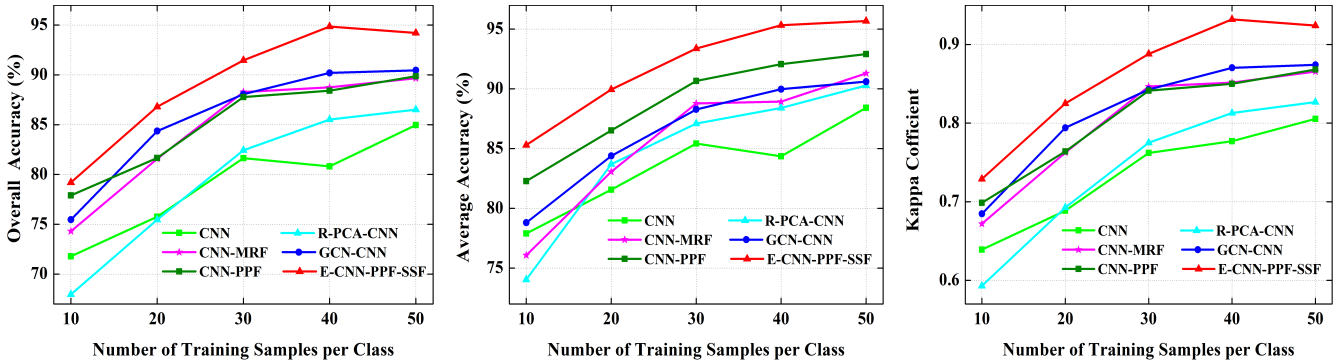


Fig. 8. OAs, AAs, and KCs of six methods with a different number of training samples per class on the UP image.

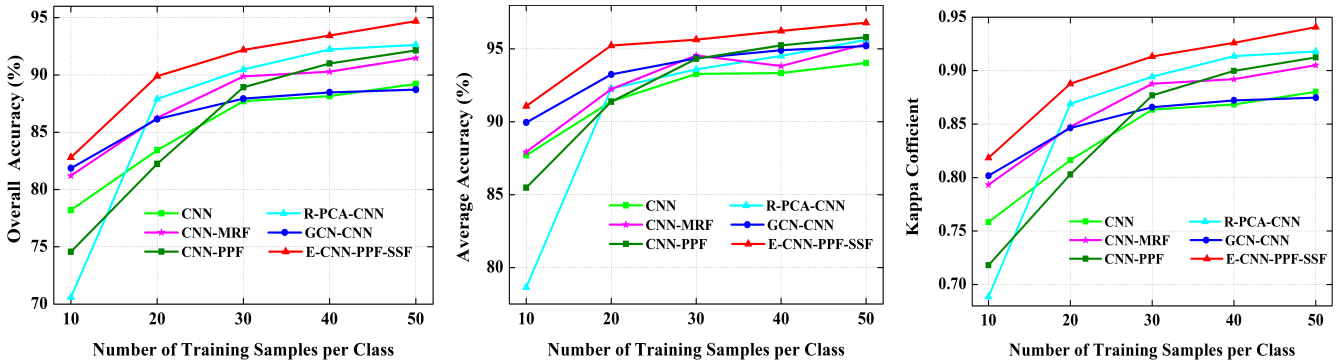


Fig. 9. OAs, AAs, and KCs of six methods with a different number of training samples per class on the SV image.

classification accuracy than the comparison approaches for the four HSI scenes. The reason can be explained as follows: 1) SSF is helpful to obtain more discriminant features; 2) sufficient training samples constructed by the PPFs are available and valuable, settling the scarcity of labeled training samples to a certain extent and guaranteeing the parameters of CNN to be fully tuned; and 3) the variance of the model is reduced and the generalization ability is improved by introducing an ensemble CNN framework, ensuring that the classification accuracy of the proposed approach is further enhanced.

V. Discussion

For the small sample problem during the process of HSI classification, we propose a deep ensemble CNN method based

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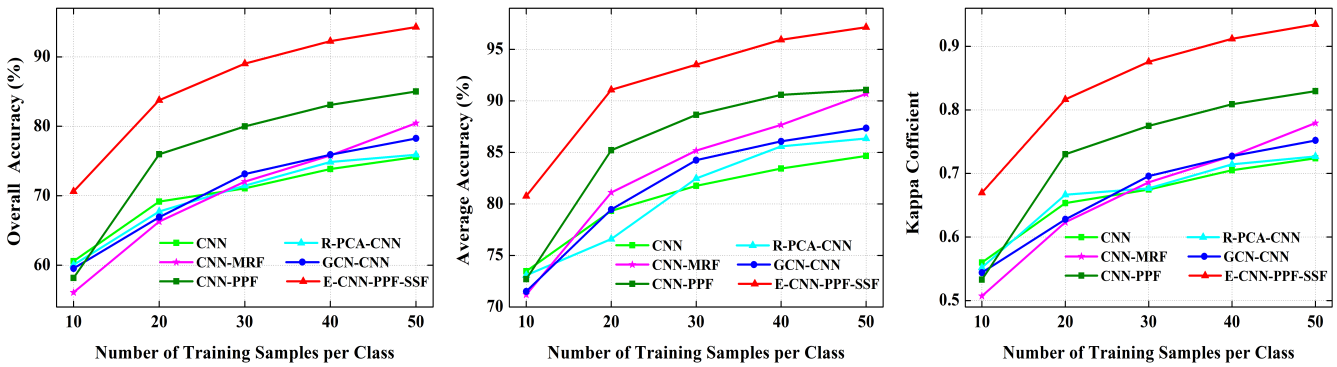


Fig. 10. OAs, AAs, and KCs of six methods with a different number of training samples per class on the IP image.

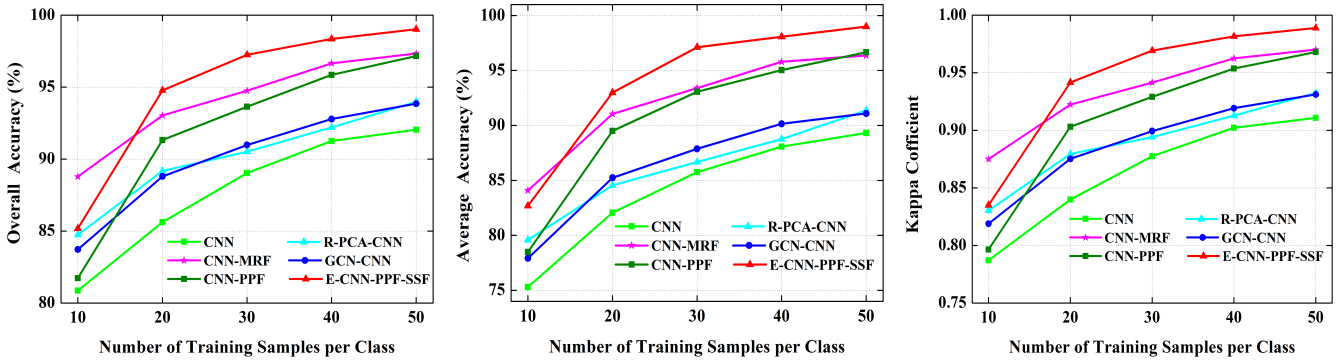


Fig. 11. OAs, AAs, and KCs of six methods with a different number of training samples per class on the KSC image.

on sample expansion in this article. Experimental outcomes in Section IV-D2 demonstrate that the studied E-CNN-PPF-SSF acquires better classification than other CNN-based deep models. Therefore, the researched E-CNN-PPF-SSF method is effective and meaningful, alleviating the problem of low classification accuracy of HSIs in the case of small samples by deep learning model to a certain extent.

In this study, we have discussed the sensitivity of ensemble size to the classification accuracy of the proposed method. According to the experimental results in Table IV, we find that, with the increase of ensemble size, the classification accuracy increases, and then, the accuracy remains at a certain level. The above trend is similar to [34] and [43]. Besides, we notice that the standard deviation decreases with the increase of ensemble size in most cases, which may explain why ensemble frameworks can make classification results stable.

On the one hand, numerous studies have verified that spectral–spatial classifiers can acquire more accurate classification results than those spectral-based classifiers since the spatial correlation of HSIs offers complementary information to spectral bands [30]. On the other hand, ensemble learning combined with multiple classifiers tends to achieve better generalization performance than any single learner and has made successful applications in HSI processing [43]. Therefore, we integrate spectral–spatial fusion and ensemble learning into the CNN-PPF model in [46]. Ablation results in Table V demonstrate that both SSF and ensemble learning contributed to the improvement of classification performance, which corresponds to the existing ideas of [30] and [43], respectively.

The approach researched in this article not only effectively alleviates the "small sample problem" faced by deep learning but also has remarkable antinoise ability. When there are noise samples in the original training set, only part of the samples containing noise can be used for training due to the application of bootstrapping. For noise-affected classifiers, the majority voting strategy can correct misclassified samples. Therefore, our ensemble framework can reduce the variance of classification model, resist the influence of noise, and improve the stability.

VI. Conclusion

To enhance the classification performance of CNN framework and alleviate the overfitting problem under the condition of small samples, a deep ensemble CNN method based on sample expansion for HSI classification is studied in this article. First, SSF is considered to provide more discriminate features. Second, the PPF algorithm is adopted to expand the number of training samples, ensuring the sufficient learning of parameters in the CNN framework. Third, deep CNN frameworks are used as subclassifiers, and the bootstrap sample technique is applied on PPFs to formulate an ensemble system with diversity. Ultimately, the majority voting strategy is applied to decide the final HSI classification result. Experimental results on the four standard HSIs demonstrate that the studied E-CNN-PPF-SSF method obtains higher classification accuracy than some conventional and advanced methods.

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Although the effectiveness of the proposed approach has been verified via ablation and comparison experiments, the prominent challenges of large memory and time consumption remain to be investigated. Therefore, adding transfer learning or other valid methods to alleviate the memory and time consumption of ensemble framework should be considered in future works.

REFERENCES

[1] G. Frati *et al.*, "Coastal sand dunes monitoring by low vegetation cover classification and digital elevation model improvement using synchronized hyperspectral and full-waveform LiDAR remote sensing," *Remote Sens.*, vol. 13, no. 1, p. 29, Dec. 2020.

[2] S. Amici and A. Piscini, "Exploring PRISMA scene for fire detection: Case study of 2019 bushfires in ben Halls gap national park, NSW, Australia," *Remote Sens.*, vol. 13, no. 8, p. 1410, Apr. 2021.

[3] Y. Han, X. Shi, S. Yang, Y. Zhang, Z. Hong, and R. Zhou, "Hyperspectral sea ice image classification based on the spectral–spatial-joint feature with the PCA network," *Remote Sens.*, vol. 13, no. 12, p. 2253, Jun. 2021.

[4] Y. Quan *et al.*, "Spectral–spatial feature extraction based CNN for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 485–488.

[5] G. Camps-Valls and L. Bruzzone, "Kernel-based methods for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 6, pp. 1351–1362, Jun. 2004.

[6] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Joint and progressive subspace analysis (JPSA) with spatial–spectral manifold alignment for semisupervised hyperspectral dimensionality reduction," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3602–3615, Jul. 2021.

[7] M. Liang, L. Jiao, S. Yang, F. Liu, B. Hou, and H. Chen, "Deep multiscale spectral-spatial feature fusion for hyperspectral images classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 8, pp. 2911–2924, Aug. 2018.

[8] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[9] X. Cao, L. Xu, D. Meng, Q. Zhao, and Z. Xu, "Integration of 3-dimensional discrete wavelet transform and Markov random field for hyperspectral image classification," *Neurocomputing*, vol. 226, pp. 90–100, Feb. 2017.

[10] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[11] J. Xia, J. Chanussot, P. Du, and X. He, "Rotation-based support vector machine ensemble in classification of hyperspectral data with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1519–1531, Mar. 2016.

[12] R. Li, Z. Pan, Y. Wang, and P. Wang, "A convolutional neural network with mapping layers for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3136–3147, May 2020.

[13] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, Jul. 2015, Art. no. 258619.

[14] Y. Chen, K. Zhu, L. Zhu, X. He, P. Ghamisi, and J. A. Benediktsson, "Automatic design of convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7048–7066, Apr. 2019.

[15] X. He, Y. Chen, and P. Ghamisi, "Heterogeneous transfer learning for hyperspectral image classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3246–3263, May 2019.

[16] L. Mou and X. X. Zhu, "Learning to pay attention on spectral domain: A spectral attention module-based convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 110–122, Sep. 2020.

[17] J. Zheng, Y. Feng, C. Bai, and J. Zhang, "Hyperspectral image classification using mixed convolutions and covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 522–534, May 2020.

[18] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[19] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[20] X. Yang *et al.*, "Synergistic 2D/3D convolutional neural network for hyperspectral image classification," *Remote Sens.*, vol. 12, no. 12, p. 2033, Jun. 2020.

[21] M. Zhu, L. Jiao, F. Liu, S. Yang, and J. Wang, "Residual spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 449–462, Jan. 2021.

[22] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[23] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[24] Z. Li, X. Cui, L. Wang, H. Zhang, X. Zhu, and Y. Zhang, "Spectral and spatial global context attention for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 4, p. 771, Feb. 2021.

[25] B. Xi *et al.*, "Multi-direction networks with attentional spectral prior for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[26] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[27] B. Liu, X. Yu, P. Zhang, A. Yu, Q. Fu, and X. Wei, "Supervised deep feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1909–1921, Nov. 2017.

[28] S. Dong, Y. Quan, W. Feng, Q. Li, G. Dauphin, and M. Xing, "Ensemble CNN based on pixel-pair and random feature selection for hyperspectral image classification with small-size training set," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2353–2356.

[29] J. Li, X. Zhao, Y. Li, Q. Du, B. Xi, and J. Hu, "Classification of hyperspectral imagery using a new fully convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 292–296, Feb. 2018.

[30] S. Dong, Y. Quan, W. Feng, G. Daunphin, L. Gao, and M. Xing, "A pixel cluster CNN and spectral–spatial fusion algorithm for hyperspectral image classification with small-size training samples," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4101–4114, 2021.

[31] M. Jiang, Y. Fang, Y. Su, G. Cai, and G. Han, "Random subspace ensemble with enhanced feature for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1373–1377, Aug. 2020.

[32] H. Jafarzadeh, M. Mahdianpari, E. Gill, F. Mohammadimanesh, and S. Homayouni, "Bagging and boosting ensemble classifiers for classification of multispectral, hyperspectral and PolSAR data: A comparative evaluation," *Remote Sens.*, vol. 13, no. 21, p. 4405, Nov. 2021.

[33] H. Su, Y. Yu, Z. Wu, and Q. Du, "Random subspace-based K-nearest class collaborative representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6840–6853, Aug. 2021.

[34] B. Waske, S. van der Linden, J. Benediktsson, A. Rabe, and P. Hostert, "Sensitivity of support vector machines to random feature selection in classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 7, pp. 2880–2889, Jul. 2010.

[35] Y. Chen, X. Zhao, and Z. Lin, "Optimizing subspace SVM ensemble for hyperspectral imagery classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1295–1305, Apr. 2014.

[36] B. Xi, J. Li, Y. Li, R. Song, W. Sun, and Q. Du, "Multiscale context-aware ensemble deep KELM for efficient hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5114–5130, Jun. 2021.

[37] X. Liu, Q. Hu, Y. Cai, and Z. Cai, "Extreme learning machine-based ensemble transfer learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3892–3902, 2020.

[38] J. Xia, P. Ghamisi, N. Yokoya, and A. Iwasaki, "Random forest ensembles and extended multiextinction profiles for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 202–216, Jan. 2018.

[39] W. Feng, W. Huang, and W. Bao, "Imbalanced hyperspectral image classification with an adaptive ensemble method based on SMOTE and rotation forest with differentiated sampling rates," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1879–1883, Dec. 2019.

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[40] W. Feng *et al.*, "Semi-supervised rotation forest based on ensemble margin theory for the classification of hyperspectral image with limited training data," *Inf. Sci.*, vol. 575, pp. 611–638, Oct. 2021.

[41] F. Lv, M. Han, and T. Qiu, "Remote sensing image classification based on ensemble extreme learning machine with stacked autoencoder," *IEEE Access*, vol. 5, pp. 9021–9031, 2017.

[42] Z. Gong, P. Zhong, J. Shan, and W. Hu, "A diversified deep ensemble for hyperspectral image classification," in *Proc. 9th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2018, pp. 1–5.

[43] Y. Chen, Y. Wang, Y. Gu, X. He, P. Ghamisi, and X. Jia, "Deep learning ensemble for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1882–1897, Jun. 2019.

[44] L. Fang, W. Zhao, N. He, and J. Zhu, "Multiscale CNNs ensemble based self-learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 9, pp. 1593–1597, Sep. 2020.

[45] X. He and Y. Chen, "Transferring CNN ensemble for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 876–880, May 2021.

[46] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[47] Q. Xu, Y. Xiao, D. Wang, and B. Luo, "CSA-MSO3DCNN: Multiscale octave 3D CNN with channel and spatial attention for hyperspectral image classification," *Remote Sens.*, vol. 12, no. 1, p. 188, Jan. 2020.

[48] Q. Lv, W. Feng, Y. Quan, G. Dauphin, L. Gao, and M. Xing, "Enhanced-random-feature-subspace-based ensemble CNN for the imbalanced hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3988–3999, 2021.

[49] J. He and X. Zhu, "Combining improved gray-level co-occurrence matrix with high density grid for myoelectric control robustness to electrode shift," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 9, pp. 1539–1548, Sep. 2016.

[50] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, Nov. 1973.

[51] R. Huang and W. He, "Using tri-training to exploit spectral and spatial information for hyperspectral data classification," in *Proc. Int. Conf. Comput. Vis. Remote Sens.*, Dec. 2012, pp. 30–33.

[52] Y. Liang, L. Wei, and Q. Lu, "Multiple feature fusion for fine classification of crops in UAV hyperspectral imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5059–5062.

[53] S. Site and S. K. Mishra, "Model for measuring accuracies of majority voting of ensemble classifier with COB and genetic algorithm," in *Proc. Int. Conf. Inf. Commun. Embedded Syst. (ICICES)*, Feb. 2013, pp. 99–103.

[54] L. Lam and C. Y. Suen, "Application of majority voting to pattern recognition: An analysis of its behavior and performance," *IEEE Trans. Syst., Man, Cybern., A, Syst. Humans*, vol. 27, no. 5, pp. 553–568, Sep. 1997.

[55] D. Ruta and B. Gabrys, "Classifier selection for majority voting," *Inf. Fusion*, vol. 6, no. 1, pp. 63–81, 2005.

[56] G. Brown and L. I. Kuncheva, "'Good' and 'bad' diversity in majority vote ensembles," in *Proc. Int. Workshop Multiple Classifier Syst.* Berlin, Germany: Springer, 2010, pp. 124–133.



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